Sprint 0

AI Toolkit Hub

5 AI-powered tools

Presented By: Aryan Patel(U01999176)
Bansari Shah(U01983468)
Laxmi Prasanna(U01950459)
Pratham Patel(U02002338)
Vivek Pitroda(U01992349)

Agenda

1. Team Introductions & Roles
2. Problem Statement & Hypothesis
3. Project Description & Solution
4. User Personas
5. Technologies & Architecture
6. Algorithms Overview
7. Project Schedule & Cadence
8. Team Working Agreement
9. Retrospective & Next Steps

Team Roles & Responsibilities

Product Owner *Aryan Patel*	Requirements, user stories, backlog priority, sprint planning
Frontend Lead *Bansari Shah*	React.js UI, Tailwind CSS, API integration, flashcard UI
Backend Lead *Pratham Patel*	Node.js APIs, JWT authentication, database, endpoints
AI/ML Lead *Vivek Pitroda*	AI API integration, prompt design, PDF to flashcard logic
QA/DevOps *Lakshmi Prasanna*	Testing, GitHub workflow, deployment, documentation

Problem Statement

Who

Students, educators, professionals, and multilingual users working with long texts, PDFs, and cross-language communication

What

Time wasted switching between multiple tools for translation, summarizing, studying, and voice translation

Why Current Solutions Fail

- Fragmented experience across 3-5 apps
- Limited history and export options
- Voice translation doesn't pair with text records

Hypothesis

We believe that building a single AI Toolkit Hub for students and multilingual professionals will reduce time spent switching tools and improve comprehension by combining language and study workflows into one unified platform.

Project Description

AI Toolkit Hub

A comprehensive web application providing five AI-powered tools for students and professionals

Language Translator

Instant text translation across 50+ languages

Text Summarizer

Condense long articles into key points

PDF to Flashcards

Automatic study material creation from PDFs

Live Voice Translator

Real-time speech translation

Text-to-Speech

Convert text-to-speech with customizable voice options

User Personas

Priya Shah

College Student, 20

Goals:

Study faster, turn lecture PDFs into revision material

Pain Points:

Too much reading, hard to identify key points

How We Help:

Summarizes chapters + generates flashcards from PDFs

Ahmed Khan

International Student, 24

Goals:

Understand English materials and communicate confidently

Pain Points:

Vocabulary barriers, slow reading, fear of misunderstanding

How We Help:

Text & voice translation with saved history

Sofia Martinez

Working Professional, 29

Goals:

Quickly understand reports/emails and create short briefs

Pain Points:

Long documents, time pressure, needs clean outputs

How We Help:

Adjustable summarizer with export options

Technologies

Frontend

- React.js
- Tailwind CSS
- Axios
- react-card-flip
- Framer Motion

Backend

- Node.js + Express.js
- Multer (file uploads)
- pdf-parse
- JWT + bcrypt

AI Services

- MyMemory Translation API
- Hugging Face (BART, FLAN-T5)
- Web Speech API

Database

- MongoDB

Algorithms & Architecture

API-based AI inference with structured workflow logic

Text Translation *MyMemory API + rule-based handling*	Input validation, language mapping, history logging
Text Summarization *facebook/bart-large-cnn*	Length control, formatting logic, response cleanup
PDF to Flashcards *google/flan-t5-large + pdf-parse*	Text extraction, chunking, Q&A generation, JSON formatting
Live Voice Translation *SpeechRecognition → Translation → Synthesis*	Real-time transcription, API chaining, error fallback
Text-to-Speech *Web Speech API*	Voice selection, pitch/rate control, playback management
Authentication *JWT + bcrypt*	Password hashing, token generation, middleware verification

Project Schedule & Cadence

1. **Sprint Length**
   Two-week sprints for adequate planning, development, and review time

2. **Sprint Planning**
   Start of each sprint - review goals and assign tasks

3. **Team Stand-ups**
   2-3 times per week - share progress and blockers

4. **Sprint Review**
   End of sprint - present completed work

5. **Sprint Retrospective**
   After review - discuss improvements for next sprint

Team Working Agreement

Communication

- Google Meet, WhatsApp, Notion
- 24-hour response time
- Early notification of unavailability

Meetings

- Saturday 4-5pm (Virtual)
- Monday 9-10pm (Virtual)
- Wednesday 4:30-5:30pm (In-person)

Work Expectations

- Complete assigned tasks by deadline
- Reach out early if blocked
- Fair workload distribution

Accountability & Quality

- Supportive accountability
- Open communication on missed deadlines
- Review work before submission

Sprint 0 Retrospective

✓ What Went Well

- Finalized project scope and features
- Confirmed tech stack and APIs
- Clear role ownership
- Effective team communication
- Updated shared wiki

→ Areas for Improvement

- Need earlier wireframes to avoid UI rework
- Consistent API response format needed before frontend integration

Concrete Actions for Sprint 1

1. Create API contract document with request/response examples for each endpoint
2. Build reusable UI component library (buttons, cards, modals, loading/error states)

Ready for Sprint 1

AI Toolkit Hub Team

Questions?